

How bias arises in Gnocchi

Peter McHale

July 17, 2026

1 Expected number of SNVs under neutrality

In a selectively neutral genomic window, the expected number of sites that are polymorphic (a site is polymorphic if an ALT allele is seen at that site in one or more members of a given cohort) is

$$\sum_i p_{c_i}(x_i) \tag{1}$$

where:

- i indexes the single-nucleotide sites in the window
- x_i is a vector of regional features (GC content, etc) associated with site i ,
- c_i is the sequence and methylation context of the site i ,
- $p_c(x)$ is the probability that a site is polymorphic, conditional on context c and regional feature vector x , and, of course, the number of individuals in the cohort.

2 Assuming uniformity of the features within a genomic window

Assume that all the features vectors $\{x_i\}$, ranging over all sites i in a window w , are approximately the same:

$$x_i = x_w.$$

Under this assumption, equation (1) simplifies to

$$\sum_i p_{c_i}(x_w) = \sum_c n_c^{(w)} p_c(x_w), \tag{2}$$

where the right side groups probabilities that appear on the left side by context, and $n_c^{(w)}$ is the number of sites in window w that have sequence and methylation context c .

3 The naive way to fit a model of the polymorphism probability

To predict the SNV count of a window with feature vector x , we need to model $p_c(x)$, the probability that a site with context c and feature vector x is polymorphic (as opposed to monomorphic) across the cohort of interest. The natural approach is to construct a parametric model of $p_c(x)$ as follows:

$$p_c(x) = \sigma(\beta_c \cdot x), \quad (3)$$

where:

- $\sigma(x)$ is the sigmoid function (bounded between 0 and 1),
- $\cdot$ is the dot product,
- β_c is a parameter vector associated with the sequence and methylation context c ,

and then train one model for each context c on an appropriate genome-wide training set of sites (each with context c , but with varying feature vectors x) that are either monomorphic or polymorphic in the cohort.

4 The training set is contaminated with selection that may correlate with x

The problem with this approach lies in choosing the training set of sites. These sites should be selectively neutral, but, in practice we don't know how to identify such sites in the noncoding genome, where functional annotations are infrequent and imprecise. Moreover, noncoding sites under selection may correlate with x (e.g., promoters, which are under selection, tend to have high GC content), causing $p_c(x)$ to be inaccurate in the tails of the x -distribution.

5 A 2-stage approach to eliminate the effect of feature-dependent contamination of the training set

De novo mutations (DNMs), by definition, have not been exposed to natural selection, and therefore inform us about the inherent mutability of a genomic site. Can we use DNMs as a training set? Not immediately: we want a neutral estimate of the polymorphism probability, which scales with cohort size, not DNM rate, which does not.

There is a way, however, to draw that DNM rate into the polymorphism probability such that it eliminates the effect of feature-dependent contamination with selection that would otherwise occur in a training set of polymorphisms

constructed from the entire noncoding genome. To leverage a set of neutrally pure DNMs, Chen et al broke the problem into two steps:

$$p_c(x) = r_c(x) p_c(\bar{x}), \quad (4)$$

where $\bar{x}$ is the average of x over all windows of interest, and

$$r_c(x) = \frac{p_c(x)}{p_c(\bar{x})}, \quad (5)$$

is a feature-dependent multiplicative adjustment to the feature-independent $p_c(\bar{x})$.

In the first step, they approximated $p_c(\bar{x})$ by the maximum-likelihood estimate:

$$p_c(\bar{x}) \approx \frac{N_c^{(p)}}{N_c^{(m)} + N_c^{(p)}} \quad (6)$$

where

- $N_c^{(m)}$ is the number of sites in the noncoding genome with context c that are monomorphic, and
- $N_c^{(p)}$ is the number of sites in the noncoding genome with context c that are polymorphic

Given the large number of training examples, and the fact that most of them have feature vectors close to $\bar{x}$, this estimate is likely to be quite accurate (though subject to low levels of contamination by selection).

In the second step, they assumed that the polymorphism probability in the cohort, $p_c(x)$, is approximately proportional to the DNM probability, $\hat{p}_c(x)$, in an arbitrary training set of DNMs and background sites (not DNMs). Under this assumption:

$$r_c(x) \approx \frac{\hat{p}_c(x)}{\hat{p}_c(\bar{x})} \quad (7)$$

where

$$\hat{p}_c(x) = \sigma(\hat{\beta}_c \cdot x), \quad (8)$$

with parameters $\hat{\beta}_c$ estimated from the training set of DNMs and background sites (not DNMs). Notice that, even though $\hat{p}_c(x)$ is sensitive to the class balance in the training set (the ratio of DNMs examples to non-DNM examples), the estimate of $r_c(x)$, which is a ratio of such probabilities, is not. Though this training set can be trusted to be neutral, even in the tails of the x -distribution, it is much smaller than that used to compute $p_c(\bar{x})$, so the estimate of $r_c(x)$ is likely to be noisier than that of $p_c(\bar{x})$ solely because of sampling noise.

6 Hypothesis: the feature-dependent adjustment is biased

As we show in our manuscript, the process of training any function of x works, in general, to minimize global bias at the expense of local bias in the tails of the x -distribution. Applied to the adjustment factor $r_c(x)$, whose training data is sparse, we expect $r_c(x)$ to be accurate near $\bar{x}$, but inaccurate (and possibly biased) in the tails (x far from $\bar{x}$). This bias would propagate, via $p_c(x)$, to the expected per-window SNV counts as a function of x , potentially explaining the bias we observe in Gnocchi at extreme GC content, for example.

7 Predictions of the hypothesis

It is instructive to consider what would happen, under the above hypothesis, as the size of the DNM training set is increased.

When the DNM training set is so small that the tails are not sampled at all, then $r_c(x)$ is expected to tend towards its value at $\bar{x}$, which is 1 by construction, and Gnocchi reduces to a context-only model, inheriting whatever bias it has as a function of x . **We performed this experiment and found not only that bias in Gnocchi was reduced, but also that it approached the levels of bias we saw in other context-only constraint metrics.**

As the DNM training set size increases, and more and more extreme values of x are sampled, $r_c(x)$ will start to capture the true function near $\bar{x}$ making Gnocchi better there, but this may come at the expense of deviating from the true function away from $\bar{x}$, in a way that we cannot predict without knowing the truth. **This is what we believe is causing the large bias we observe in Gnocchi.**

Eventually, if enough DNM training data existed to densely sample extreme x , then we would expect the bias at extreme x to disappear. **We approximated this regime by increasing the number of background sites (only) in the DNM training set, retrained $r_c(x)$, and observed that bias below the level we reported in our paper.**

Together these experiments lend support to our hypothesis that the sparseness of the DNM training set is responsible for the large bias observed in Gnocchi as a function of GC content.

8 A proposal for Gnocchi 2.0

Though DNMs ameliorate the effect of selection contamination of the training set, the trade-off is that the sparseness of DNMs introduces bias into the constraint metric, precisely where it was hoped DNMs would help. Perhaps the quest to avoid contamination of the training set with sites under selection was misguided, and it is better to tolerate some contamination if it yields a less biased constraint score where it matters most. This suggests an approach to a possibly improved constraint score: capture the x dependence in Eq. (3) by training directly on

dense gnomAD data, instead of factorizing it and training one of the factors on sparse DNM data. Since this requires a new training process, we deemed this experiment out-of-scope for this rebuttal.